Advance - CSS

infomax
Computer Academy

CSS Part 2

CSS – Cascading Style Sheet (Advance)

CSS Gradients

CSS gradients let you display smooth transitions between two or more specified colors.

CSS defines three types of gradients:

- Linear Gradients (goes down/up/left/right/diagonally)
- Radial Gradients (defined by their center)
- Conic Gradients (rotated around a center point)

Linear Gradients

To create a linear gradient you must define at least two color stops. Color stops are the colors you want to render smooth transitions among. You can also set a starting point and a direction (or an angle) along with the gradient effect.

Syntax:

background-image: linear-gradient(direction, color-stop1, color-stop2, ...);

Applying Gradient

- Direction - Top to Bottom (this is default)

  background-image: linear-gradient(red, yellow);

- Direction - Left to Right

  background-image: linear-gradient(to right, red , yellow);

- Direction – Diagonal

  background-image: linear-gradient(to bottom right, red, yellow);

- Using Angles

  background-image: linear-gradient(180deg, red, yellow);

- Using Multiple Color Stops

  background-image: linear-gradient(red, yellow, green);

Applying Gradient Cont..

- Using Transparency
  background-image: linear-gradient(to right, rgba(255,0,0,0), rgba(255,0,0,1));
- Repeating a linear-gradient
  background-image: repeating-linear-gradient(red, yellow 10%, green 20%);

CSS Radial Gradients

A radial gradient is defined by its center.

To create a radial gradient you must also define at least two color stops.

Syntax:

background-image: radial-gradient(shape size at position, start-color, ..., last-color);

Example:

```
#grad {
background-image: radial-gradient(red, yellow, green);
}
```

Applying CSS Radial Gradients

- Radial Gradient - Evenly Spaced Color Stops (this is default)

  background-image: radial-gradient(red, yellow, green);

- Radial Gradient - Differently Spaced Color Stops

  background-image: radial-gradient(red 5%, yellow 15%, green 60%);

- Set Shape

  background-image: radial-gradient(circle, red, yellow, green);

- Repeating a radial-gradient

  background-image: repeating-radial-gradient(red, yellow 10%, green 15%);

Applying CSS Radial Gradients Conti.

Use of Different Size Keywords

The size parameter defines the size of the gradient. It can take four values:

- closest-side farthest-side
- closest-corner farthest-corner
- Example 1:
  background-image: radial-gradient(closest-side at 60% 55%, red, yellow, black);
- Example 2:
  background-image: radial-gradient(farthest-side at 60% 55%, red, yellow, black);

CSS Conic Gradients

A conic gradient is a gradient with color transitions rotated around a center point.

To create a conic gradient you must define at least two colors.

Syntax:

background-image: conic-gradient([from angle] [at position,] color [degree], color [degree], ...);

Example:

background-image: conic-gradient(red, yellow, green);

Applying CSS Conic Gradients

- Conic Gradient multiple colors

  background-image: conic-gradient(red, yellow, green, blue, black);

- Conic Gradient with Colors and Degrees

  background-image: conic-gradient(red 45deg, yellow 90deg, green 210deg);

  background-image: conic-gradient(red 0deg, red 90deg, yellow 90deg, yellow 180deg, green 180deg, green 270deg, blue 270deg);

- Conic Gradient With Specified From Angle

  background-image: conic-gradient(from 90deg, red, yellow, green);

- Conic Gradient With Specified Center Position

  background-image: conic-gradient(at 60% 45%, red, yellow, green);

Applying CSS Conic Gradients Conti.

- Repeating a Conic Gradient

    background-image: repeating-conic-gradient(red 10%, yellow 20%);

CSS Web Fonts

In the @font-face rule; first define a name for the font (e.g. myFirstFont) and then point to the font file.

To use the font for an HTML element, refer to the name of the font (myFirstFont) through the font-family property:

```
@font-face {
  font-family: myFirstFont;
  src: url(sansation_light.woff);
}

div {
  font-family: myFirstFont;
}
```

CSS 2D Transforms

CSS transforms allow you to move, rotate, scale, and skew elements.

Mouse over the element below to see a 2D transformation:

With the CSS transform property you can use the following 2D transformation methods:

- translate()
- rotate()
- scaleX()
- scaleY()
- scale()
- skewX()
- skewY()
- skew()
- matrix()

The translate() Method

The translate() method moves an element from its current position (according to the parameters given for the X-axis and the Y-axis).

```
div:hover {
  transform: translate(50px, 100px);
}
```

The rotate() Method

The rotate() method rotates an element clockwise or counter-clockwise according to a given degree.

```
div:hover {
  transform: rotate(20deg);
}
```

The scale() Method

The scale() method increases or decreases the size of an element (according to the parameters given for the width and height).

```
div :hover{
  transform: scale(2, 3);
}
```

The scaleX() Method

The scaleX() method increases or decreases the width of an element.

```
Div:hover {
  transform: scaleX(2);
}
```

The scaleY() Method

The scaleY() method increases or decreases the height of an element.

```
div:hover {
  transform: scaleY(3);
}
```

The skewX() Method

The skewX() method skews an element along the X-axis by the given angle.

```
div:hoevr {
  transform: skewX(20deg);
}
```

The skewY() Method

The skewY() method skews an element along the Y-axis by the given angle.

```
div:hover {
  transform: skewY(20deg);
}
```

The skew() Method

The skew() method skews an element along the X and Y-axis by the given angles.

```
div:hover {
  transform: skew(20deg, 10deg);
}
```

If the second parameter is not specified, it has a zero value. So, the following example skews the <div> element 20 degrees along the X-axis:

```
div {
  transform: skew(20deg);
}
```

The matrix() Method

The matrix() method combines all the 2D transform methods into one.

The matrix() method take six parameters, containing mathematic functions, which allows you to rotate, scale, move (translate), and skew elements.

The parameters are as follow: matrix(scaleX(), skewY(), skewX(), scaleY(), translateX(), translateY())

```
div {
  transform: matrix(1, -0.3, 0, 1, 0, 0);
}
```

CSS 3D Transforms

With the CSS transform property you can use the following 3D transformation methods:

- rotateX()
- rotateY()
- rotateZ()

he rotateX() Method

The rotateX() method rotates an element around its X-axis at a given degree:

```
#myDiv {
  transform: rotateX(150deg);
}
```

The rotateY() Method

The rotateY() method rotates an element around its Y-axis at a given degree:

```
#myDiv {
  transform: rotateY(150deg);
}
```

The rotateZ() Method

The rotateZ() method rotates an element around its Z-axis at a given degree:

```
#myDiv {
  transform: rotateZ(90deg);
}
```

CSS Transitions

CSS transitions allows you to change property values smoothly, over a given duration.

Transition Properties:

- transition
- transition-delay
- transition-duration
- transition-property
- transition-timing-function

Example:

```
div {
  width: 100px;
  height: 100px;
  background: red;
  transition: width 2s;
}
div:hover {
  width: 300px;
}
```

Change Several Property Values

```
div {
  width: 100px;
  height: 100px;
  background: red;
  transition: width 2s, height 4s;
}
div:hover {
  width: 300px;
  height: 300px;
}
```

THANK YOU

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